

1. General Description

The EMS3157-Q100 is a single SPDT low on-resistance analog switch. It can operate from 1.5V to 5.5V power supply. The device offers low ON-state resistance and excellent ON-state resistance matching with break-before-make feature, to prevent signal distortion during the transferring of a signal from one channel to another. The device is capable of truly isolation. Even when A overrides V_{CC} , very little current will flow back to the supply.

This product has been qualified to the Automotive Electronics Council (AEC) standard Q100 (Grade 1) and is suitable for use in automotive applications

2. Features and Benefits

- Qualified for Automotive Applications
- Wide Supply Range from 1.5 V to 5.5 V
- Low ON-state resistance, typically 1.9Ω ($V_{CC}=5.5 \text{ V}$)
- 1.8 V Compatible Control Logic Inputs
- High Off-Isolation: -80 dB @ 10 kHz
- Low Channel-to-Channel Crosstalk: -82 dB @ 10 kHz
- High Bandwidth: -3dB @ 800MHz
- Low Quiescent Current ($< 1\mu\text{A}$) when $V_{CC} = 3.3 \text{ V}$
- Latch-up performance exceeds 200 mA
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3000 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V

3. Applications

- Audio, Video, UART, USB2.0 Signal and Supply Routing
- Cell phones and TWS headset

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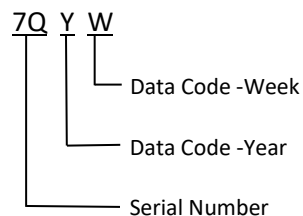
4. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EMS3157GW-Q100	7Q YW	SOT363	SOT363 package, 6 pins 2.1 mm × 1.25 mm; 1.1 mm (Max) height	3000

MARKING INFORMATION

NOTE: YW = Data Code.



5. Disclaimer

EnergyMath Corp reserves the right to make any change in circuit design, or specifications without prior notice.

6. Pinning Information

6.1. Pin map

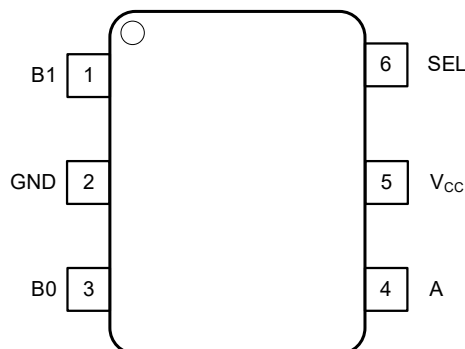


Fig 1. Top view pin configuration SOT363

6.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
B1	1	Analog/Digital Signal Port (Normally open)
GND	2	Ground (0V)
B0	3	Analog/Digital Signal Port (Normally closed)
A	4	Common Signal Port
V _{CC}	5	Supply voltage
SEL	6	Logic Input Control

7. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level.

Input SEL	Channel on
L	B0
H	B1

8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	supply voltage		-0.3	6.5	V
V _{SEL}	control Input Voltage		-0.3	6.5	V
V _A	signal input voltage		-0.3	6.5	V
I _{max}	continuous Current Through A, B0, B1			±100	mA
	peak Current Through A, B0, B1 (pulsed at 1ms 50% duty cycle)			±200	mA
T _J	junction Temperature under Bias			150	°C
T _S	storage temperature		-65	150	°C
T _L	lead Temperature (Soldering, 10 seconds)			260	°C
R _{θJA}	thermal resistance			350	°C/W

9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V _{CC}	supply voltage		1.5	5.5	V
V _{SEL}	control input voltage		0	5.5	V
V _A	signal input voltage		0	V _{CC}	V
T _{amb}	ambient temperature		-40	125	°C

10. Electrical Characteristics

Table 6. Static characteristics

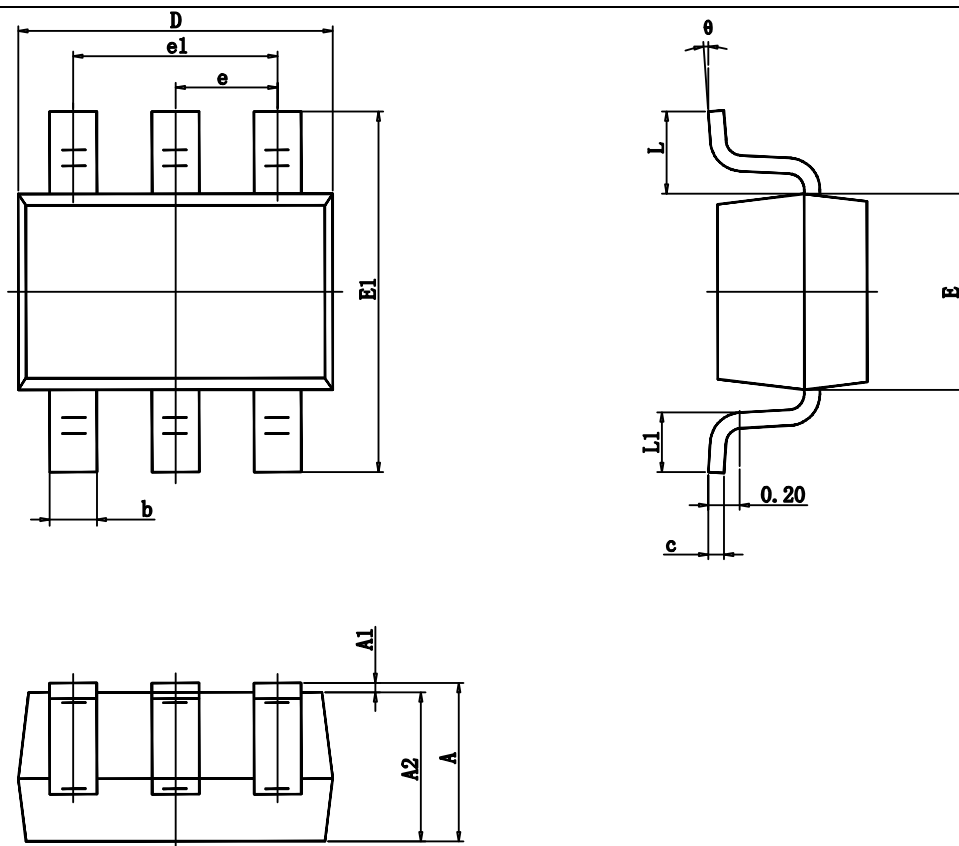
$V_{CC}=3.3\text{ V}$, unless otherwise specified. All typical values are measured at $T_{amb} = 25\text{ }^{\circ}\text{C}$.

Symbol	Parameter	Conditions	-40°C to +85°C			-40°C to +125°C		Unit
			Min	Typ	Max	Min	Max	
DC Electrical Characteristics								
V _{IH}	input logic high level	V _{CC} = 3.4 V to 5.5 V	1.6			1.6		V
		V _{CC} = 1.5 V to 3.3 V	1.4			1.4		V
V _{IL}	input logic low level	V _{CC} = 3.4 V to 5.5 V			0.6		0.55	V
		V _{CC} = 1.5 V to 3.3 V			0.35		0.3	V
I _{CC}	supply current	I _A = 0, V _{SEL} = 0 or V _{SEL} = V _{CC}			1.0		1.0	μA
ΔI _{CC}	additional supply current	I _A = 0, V _{CC} = 2.5 V or 4.2 V V _{SEL} = 1.6 V			3.5		3.5	μA
I _{off}	OFF-state switch leakage current from A to B0 (or B1)	V _A = 5 V , V _{B0} or V _{B1} = 0 V			±2.0		±2.0	μA
R _{ON}	on-resistance	V _{CC} = 3.5 V , V _A = 0.8 V to 2.0 V , I _A = 10 mA		3.4	5.0		5.5	Ω
		V _{CC} = 4.5 V , V _A = 0.8 V to 2.3 V , I _A = 10 mA		2.4	3.5		4.0	Ω
		V _{CC} = 5.5 V , V _A = 0.8 V to 2.6 V , I _A = 10 mA		1.9	3.0		3.5	Ω
R _{FLAT}	on-resistance flatness	V _{CC} = 3.5 V , V _A = 0.8 V to 2.0 V , I _A = 10 mA		0.2	0.6		0.65	Ω
		V _{CC} = 4.5 V , V _A = 0.8 V to 2.3 V , I _A = 10 mA		0.25	0.6		0.65	Ω
		V _{CC} = 5.5 V , V _A = 0.8 V to 2.6 V , I _A = 10 mA		0.3	0.7		0.72	Ω
ΔR _{ON}	on-resistance matching between channels	V _{CC} = 3.5 V , V _A = 0.8 V to 2.0 V , I _A = 10 mA		0.1	0.5		0.5	Ω
		V _{CC} = 4.5 V , V _A = 0.8 V to 2.3 V , I _A = 10 mA		0.1	0.3		0.32	Ω
		V _{CC} = 5.5 V , V _A = 0.8 V to 2.6 V , I _A = 10 mA		0.1	0.2		0.25	Ω

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Symbol	Parameter	Conditions	-40°C to +85°C			-40°C to +125°C		Unit
			Min	Typ	Max	Min	Max	
AC Electronics Characteristics								
t _{ON}	turn-On Time	V _A =1.5V, C _L =50pF, R _L =50Ω		470	600		650	ns
t _{OFF}	turn-Off Time	V _A =1.5V, C _L =50pF, R _L =50Ω		120	200		250	ns
t _{BBM}	break-before-make time	V _A =1.5V, C _L =50pF, R _L =50Ω		600	1000		1100	ns
BW	-3dB Bandwidth	C _L =0pF		800				MHz
OIRR	off isolation	f=10 KHz		-80				dB
		f=240 MHz		-31				dB
X _{TALK}	crosstalk	f=10 KHz		-82				dB
		f=240 MHz		-26				dB
THD	total harmonic distortion	f=20Hz to 20KHz V _A =600mVpp, R _L =32Ω		-80				dB
Capacitance								
C _{ON}	On Capacitance	f=100KHz		12				pF
C _{OFF}	Off Capacitance	f=100KHz		4				pF

11. Package Outline

SOT363


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	0.900	1.100	0.035	0.043
A1	0.000	0.100	0.000	0.004
A2	0.900	1.000	0.035	0.039
b	0.150	0.350	0.006	0.014
c	0.110	0.175	0.004	0.007
D	2.000	2.200	0.079	0.087
E	1.150	1.350	0.045	0.053
E1	2.150	2.450	0.085	0.096
e	0.650 TYP.		0.026 TYP.	
e1	1.200	1.400	0.047	0.055
L	0.525 REF.		0.021 REF.	
L1	0.260	0.460	0.010	0.018
theta	0°	8°	0°	8°

12. Tape and Reel Information

12.1. Carrier tape dimensions

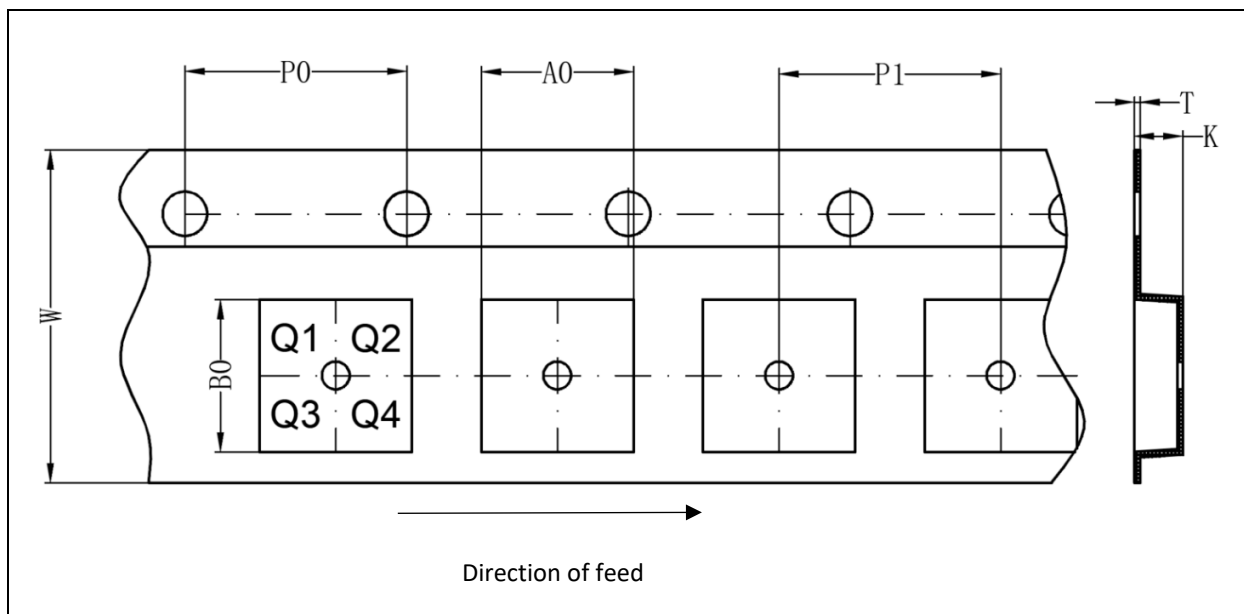


Table 7. Carrier tape dimensions

Type number ending	Package version	A0(mm)	B0(mm)	K(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
GW	SOT353	2.55	2.55	1.2	0.2	4	8	4	Q3

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12.2. Reel and box dimensions

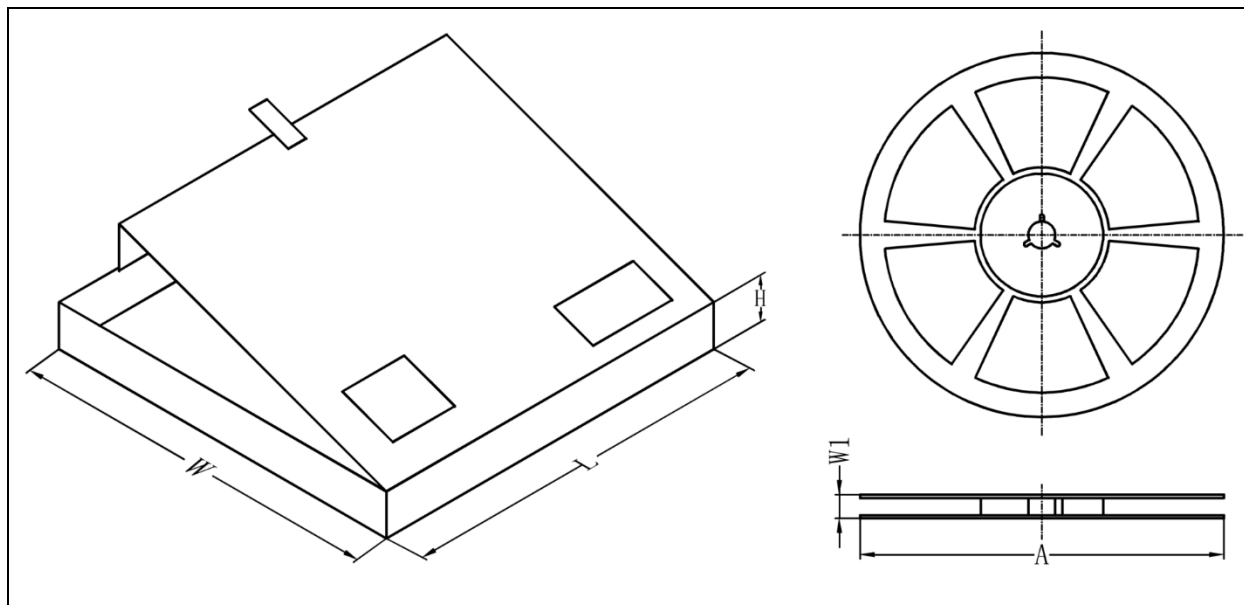


Table 8. Dimensions and quantities

Type number ending	Package version	MSL	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
GW	SOT353	MSL1	180	12.2	3000	1	210x200x40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

13. Revision History

Table 9. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EMS3157-Q100 Rev. 0.92	Nov 25, 2025	Draft datasheet		